

# Robust Statistics: An Overview

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## Introduction

A robust statistic is one that remains useful when the data deviate from the idealised model – heavy tails, outliers, skewed contamination. Classical estimators like the mean and SD are efficient under normality but break down under even modest contamination. Robust alternatives sacrifice a little efficiency for dramatic stability, and are the default in many applied fields where data quality cannot be guaranteed.

## Prerequisites

The reader should know the mean, median, SD, and MAD. Familiarity with outlier concepts helps.

## Theory

Two key concepts from robustness theory:

**Breakdown point.** The largest fraction of contaminated observations an estimator can tolerate before giving an arbitrary answer. The arithmetic mean has breakdown 0: a single extreme point can push it arbitrarily far. The median has breakdown 0.5: up to half the sample can be contaminated before the median is affected.

**Influence function.** Measures the infinitesimal effect of adding a single observation at  $x$  on the estimator. The arithmetic mean has a linear, unbounded influence function; the median has a bounded (step-shaped) influence function.

Robust estimators aim for high breakdown point and bounded influence function, typically with moderate loss of efficiency at the normal model.

**Robust alternatives** to classical summaries:

| Classical | Robust                                                      | Breakdown | Efficiency at normal |
|-----------|-------------------------------------------------------------|-----------|----------------------|
| Mean      | Median                                                      | 0.5       | 64%                  |
| Mean      | 20% trimmed mean                                            | 0.2       | ~95%                 |
| Mean      | Huber M-estimator                                           | 0         | ~95%                 |
| SD        | MAD                                                         | 0.5       | 37%                  |
| SD        | Qn, Sn                                                      | 0.5       | ~82%                 |
| OLS       | Robust regression ( <code>r1m</code> , <code>lmrob</code> ) | 0 to 0.5  | 95%                  |

## Assumptions

Robustness is an explicit relaxation of distributional assumptions; it asks how the estimator behaves under deviation, not whether deviation is present.

## R Implementation

```
library(robustbase)
library(MASS)
```

```

set.seed(2026)
clean <- rnorm(100, mean = 50, sd = 10)
contaminated <- c(clean[1:95], rnorm(5, mean = 200, sd = 5))

classical <- c(mean = mean(contaminated), sd = sd(contaminated))
robust <- c(median = median(contaminated),
           mad = mad(contaminated),
           Qn = Qn(contaminated),
           Sn = Sn(contaminated))

rbind(classical = c(classical, NA, NA), robust)

# Robust regression example
n <- 80
x <- rnorm(n)
y <- 2 + 1.5 * x + rnorm(n)
y[c(5, 10, 15)] <- y[c(5, 10, 15)] + c(15, -15, 20) # inject outliers

coef(lm(y ~ x))
coef(rlm(y ~ x))           # M-estimator
coef(lmrob(y ~ x))         # MM-estimator with high breakdown

```

## Output & Results

|  | mean  | sd   | median | mad   | Qn    | Sn    |
|--|-------|------|--------|-------|-------|-------|
|  | 57.84 | 32.9 | 50.28  | 10.21 | 10.45 | 10.72 |

Classical mean is pulled to 57.8 by the five extreme values; every robust estimator remains near the true 50. For the regression: OLS slope is 1.23 (pulled by outliers); `rlm` gives 1.47 (close to truth of 1.5); `lmrob` gives 1.48.

## Interpretation

Use robust statistics as the default in biomedical work where outliers or heavy tails are typical. A robust summary plus a classical one can be reported side by side; divergence between them signals contamination that warrants investigation.

## Practical Tips

- MM-estimators (`lmrob`) combine high breakdown with high efficiency and are the modern robust-regression default.
- Robust methods are not a license to ignore outliers; always investigate before reporting.
- For heavy-tailed data, a Student-t likelihood (`brms` or `heavy` package) is an alternative to robust M-estimators.
- Robust confidence intervals are usually narrower than a non-robust CI on contaminated data, reflecting the estimator's stability.
- Specialised packages: `robustbase`, `WRS2` (Wilcox), `quantreg` (quantile regression), `MASS::rlm`.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.

- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Descriptive statistics and Table 1